

Pythagoras, Fibonacci, DNA and me: Is there a Formula for Life?

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".../...What is interesting, that my colleague mathematician Perez thinks, is that there is a kind harmonization of sequences of genetic information which is carried both by our own chromosomes than by viruses. So, Nature does not accept just anything, that is to say we can do anything to Nature. But if you make an Artificial construction, it has little chances of survival. That is to say that Nature loves harmonious constructions and what is foreigner for example is not well tolerated.../..." Excerpt from the historic interview with Luc Montagnier affirming the laboratory-manipulated origin of the SARS-CoV2 virus (CNews TV broadcast of April 17, 2020)

This is the story of 35 years of research on the biomathematics of DNA, genes and genomes. Paradoxically, it all started in the 1980s with research... In Artificial Intelligence! More precisely on self-organization, this "sub-AI" which emerged from Artificial neural networks. It is the wonder at the self-organization of the game of life or Conway's automaton: how a little "U" forms of 7 binary cells, springing from chance, will generate this "clown" face rich in more than a hundred cells? Then my network **FRACTAL CHAOS** was born, a fusion of this theory of the autonomy of living systems Varela autopoiesis where each neuron will be a fractal close to Mandelbrot fractals. This real network delocalized holographic memory has the memory of "déjà vu". Contemporary of the American pioneers of Neural Networks Nobel Prize winners Hopfield and Hinton, I published in 1988 **"New Paths to AI" ("De nouvelles voies vers l'intelligence Artificielle")**. It is by trying to understand the hypersensitivity of **FRACTAL CHAOS** in a specific region of the fractal $x(n+1) = 1 - \mu \cdot x(n)^2$ that I discover that this point when $\mu=1$ is the value of the Golden Ratio! Indeed **$0.618=1-(0.618)^2$** We are in 1990, the HUGO project for sequencing the human genome begins, I then have the intuition to search for Fibonacci proportions between TCAG bases in known DNA sequences. We know in fact that the ratio of 2 consecutive Fibonacci numbers tend towards Phi the golden ratio.

Fibonacci: 1 1 2 3 5 8 13 21 34 55 89 144 233...

This discovery published in 1997 in the book **"DNA decyphered" ("L'ADN décrypte")** for example, continues to be explored for 35 years, correlated with cancer mutations in our mtDNA genome mitochondrial or, in 2020 with RIP my friend Luc Montagnier to illustrate how the Sarscov2 genome, although manipulated by man, will have been able to mutate to find the Laws of Harmony of Nature on his own... Likewise in 2019, in the peer review article "Epigenetics Theoretical Limits of Synthetic Genomes". We demonstrate how the Fibonacci TC/AG resonances of the real Caulobacter genome are completely destroyed in its synthetic genome which only retains the genes by removing junk DNA. The same goes for the genome synthetic Yeast yeast from the American Craig Venter. These last few days I am still amazed by discovering the strong link between the key to geometry, the triangle of Pythagoras $a^2+b^2=c^2$, and that of harmony, Fibonacci: If n is the rank of a Fibonacci number: **$(F_n)^2 + F_{(n+1)}^2 = F_{n+5}$** . Example $(3)^2 + (5)^2 = 34$. Ranks 4 5 and 9 in the Fibonacci sequence above. Among our ancients, "square numbers" represented surfaces (2d), while here the diagonal of the right triangle remains a line (1d) which relates to the addition of 2 surfaces (2d) all in the harmony of 3 Fibonacci numbers 3 5 and 34. Imagine a Pentagon or 5-pointed star inscribed in a circle. $F(n)$ and $F(n+1)$ are the first 2 vertices, $F(n+5)$ will then coincide with the first vertex after a spiral turn of 2π . We have the beginning of a golden spiral: 3^2 , 5^2 and 34 for $n=4$. This first complete turn strongly resembles the octave change in Music...

In 2009 with the book **CODEX BIOGENESIS**, then in this 2010 article "Codon populations in single-stranded whole human genome DNA Are fractal and fine-tuned by the Golden Ratio 1.618", We have accumulated in successive partitions obeying the famous fractal called "Dragon Curve" the billion triplet codons constituting the entire human genome. These successive partitions were done with for partition key matrix the table of 64 codons of the universal genetic code. For example let's cod column T and C on the one hand and codons from columns A and G on the other hand. We then obtained the perfect balance of 1 (equality). On the contrary if we compared the codons of columns T and A on the one hand and C and G on the other hand we obtained this curious ratio linked to the Golden Ratio **$(3-\Phi)/2 = 0.690983$** .

And so on alternately between these 2 attractors for the other successive partitions. We have calls this last ratio HGO for Human Genome Optimum. We then demonstrate in several articles of 2018 on breast cancer, prostate cancer, glioblastoma, neuroblastoma, such as "Towards a Universal Law Controlling all Human Cancer Chromosome LOH Deletions, Perspectives in Prostate and Breast Cancers" how deletions associated with one type of cancer in a single Chromosome will negatively affect and unbalance HGO on a genome-wide scale. This genome imbalance then risking entering a spiral of disorder and decline on the scale of the integrity of this fragile balance: HGO.

Everything that follows is partially described in the book **CODEX BIOGENESIS**, then completely in the article by 2021 **"Six fractal codes of Life... /..."** preface by my friend Luc Montagnier. In 1997 it was out of pure aesthetic sense that I found it abnormal that Mother Nature used 3 languages too heterogeneous as DNA, RNA and amino acids to create Life. Why changes the base T of DNA to basis U of RNA? Why not the slightest link in atomic composition between the 64 RNA codons and the 20 amino acids? Rather than asking myself like the others the classic question of the genesis of life: RNA or amino acids? , I decided that there might be some sort of common metacode that would unify the 3 DNA, RNA and protein universe. Was there a common denominator? Yes, the atomic mass... and, perhaps also... Geometry! **"Where there is matter, there is geometry."** **Johannes Kepler**. For me geometry meaning "projection" like the cosine or like these cathedral builders began by studying the shadow of a vertical mast pole shadow projected on the ground...

The Formula for Life:

I then look for a formula for digital projection of atomic masses, geometry, it would inherit from Pi, the key to trigonometry dear to Pythagoras, and, of harmony, it would inherit Phi, the Golden Ratio associated with the Fibonacci sequence. In 1997, I discovered this formula:

Proj(m) = residues [1 - 4. Pi. (root fi). fi. (fi*2)] x m With "residues" = the decimal remainder of the result, (Exp. 345 for 5.345) and $fi = 0.618 = 1/\Phi$, $\Phi=1.618$ the golden ratio. Strangely then, for DNA, RNA, amino acid or any biochemical compound of atomic mass m, this projection leads to a scale of integers multiples of $\pi/10$ and regularly positioned on the linear scale: - $3\pi/10$, - $2\pi/10$, $0\pi/10$, $+1\pi/10$, $+2\pi/10$, $+3\pi/10$... etc... We can then ENCODE any nucleotide component T U C A G, DNA, RNA, triplet codon, amino acid etc... following this common numerical scale of integers -2 -1 0 +1 +2 +3... From this "Formula for life" will emerge in cascade multiple levels of interlocking codes ranging from the scale of isotopes down to the level of standing waves (frequencies) at the scale of entire chromosomes. A first surprise is the fact that this projection formula works better for the average atomic masses of atoms than for a single isotope. For example the average oxygen atom is a weighted mix O16 O17 O18. The formula works well better for this average oxygen which is that of our atmosphere than for deuterium alone O18.

The Master Code of Biology:

For any double strand of DNA, we can therefore code this sequence by assigning to each codon the resulting number of his projection formula for life. The same is true for the amino acids corresponding to these codons. So suites of codes that we call Genomics (DNA codes) and Proteomics (amino acid codes). Pour each position, we can also calculate a code which corresponds to the codes of the sequence seen from this position, a kind of radiation of the sequence in this position, in the mathematical sense, it is a kind of integral. We obtain therefore for genomics and proteomics 2 sequences of "radiations" which we can even classify and order from the most influential at least influential. Then 2 sequences of classification of positions reflecting the influence of each position on the rest of the sequence. We then discover that these 2 curves said of the Master Code are always strongly correlated (often more than 95%) between the Genomics and Proteomics signatures! We can thus verify the reality of Master Code, unifying the DNA sequences and amino acids corresponding to them in 2 correlated signatures. And this is true for genes allowing the localization of functional or mutagenic sites but also for chromosomes integers of which only a few % are known to code for proteins. This is even verified for an imaginary gene which would be produced by exploring the 64 codons of the universal genetic code table by column: we obtain a signature in "M", whose rising branches therefore have little influence in the classification would be those of column 1(T) and 3(A) and whose descending branches, therefore very influential, would be columns 2(C) and 4(G). This comes from the fact that the electrical nature of DNA is ensured by the sole bases G of the double strand, therefore C or G of the single strand (C facing has a G on the complementary strand). Fractal Textures, waves and karyotypes: on the 2 Genomics and Proteomics Curves, when we analyze their texture between 2 successive points (n, n+1) then the cumulons (1 if increasing slope between 2 successive points and 0 if slope decreasing,

which is equivalent to making a first order derivative) we obtain a strange result: for Genomics the ratios increasing/decreasing slopes are distributed in Gauss type dispersion around a value of 61% (0.618 number gold). On the contrary for Proteomics, there is a sort of modulation happening centered on 2 attractors 61% and 30.5%, (0.618/2 and 0.618). It is therefore a real Binary code! If we now look at slopes of order 2 ($n, n+2$) then of order 3 ($n, n+3$) etc... It then appears by Genomics of real WAVES of periods 3, 4, 5...etc... On the contrary for Proteomics, the structures are erratic and not periodic. So the combination of these Genomics waves and the Proteomics Binary code applied to the 23 chromosomes humans will be completely correlated with the famous Karyotypes with the characteristic alternating black gray white bands of human genetic heritage. When we know that these Karyotypes are produced by interferometry, we have here a confirmation of the WAVE nature of our genome. And we have just discovered the theoretical laws. Generalized standing waves: until now the sequences were segmented by codon triplets of 3 bases, it is Genomics waves are already emerging.

Let's generalize the segmentation of sequences by quadruplets, quintuplets... N-tuplets. We thus obtain, for any sequence, digital standing waves including for entire chromosomes. By example in the 2018 article "Intriguing Humans and Primates chromosomes 4", We demonstrate for 6 primates (Homosapiens, Neanderthal, Chimpanzee, Orangutan, Gorilla and Macaque) whose genes are known to be very similar, enormous differences in standing waves which, moreover, have periods in Fibonacci numbers! Genomics period of 21 base pairs for the chimpanzee and urangutan, 34 base pairs for man, and 55 base pairs for the gorilla!

A triple proof to conclude: 1- we can here the usefulness of all this non-coding DNA called JUNKDNA. 2- This integrity of chromosomes is materialized by digital standing waves. 3- These waves are for these primate chromosomes... Fibonacci numbers... We have just proven here that these 95% of the DNA of our genome, clumsily named JUNK DNA because its function was unknown, guarantees INTEGRITY and COHESION biomathematics and waves of our genetic heritage. Dare to modify it, or even patent it parties (what is in progress) is a crime against Humanity! The human genome is not a dustbin of applied science.

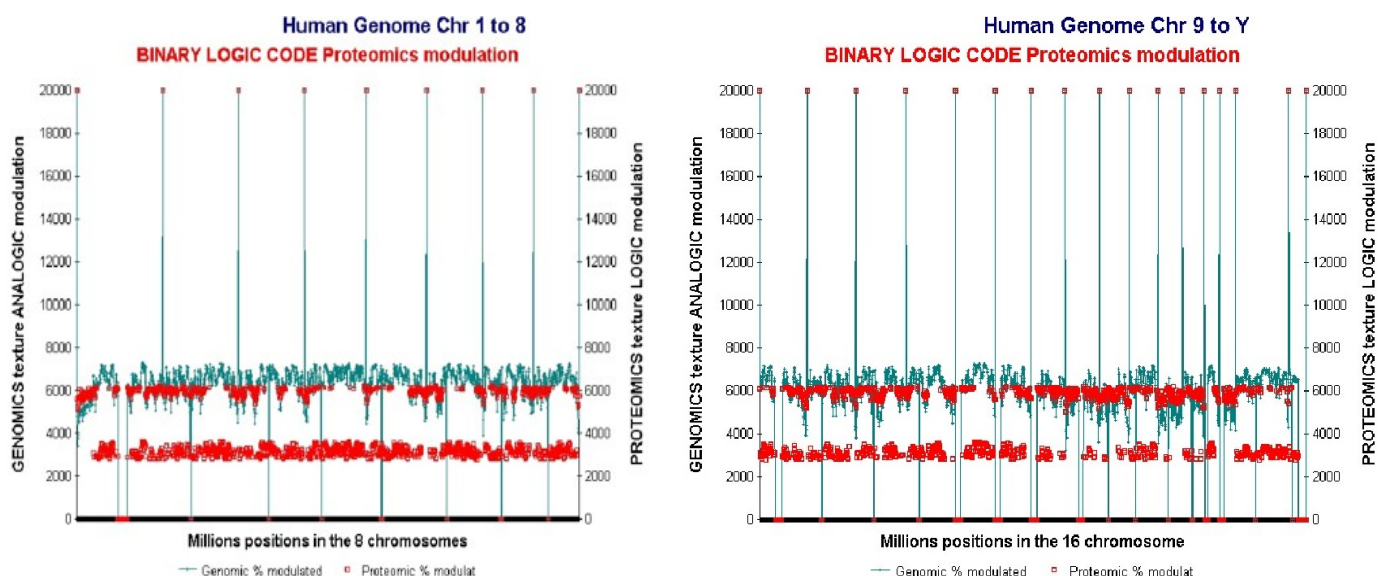


Figure 16 The 16 other remaining Chromosomes (1 to 8) of the whole HUMAN GEN

Main REFERENCE

Perez, J. C., & Montagnier, L. (2021). SIX FRACTAL CODES OF LIFE FROM BIOATOMS ATOMIC MASS TO CHROMOSOMES NUMERICAL STANDING WAVES: THREE BREAKTHROUGHS IN ASTROBIOLOGY, CANCERS AND ARTIFICIAL INTELLIGENCE. *International Journal of Research -GRANTHAALAYAH*, 9(9), 133–191. <https://doi.org/10.29121/granthaalayah.v9.i9.2021.4191>

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